A Two-Layer Encoding Learning Swarm Optimizer Based on Frequent Itemsets for Sparse Large-Scale Multi-Objective Optimization

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***Abstract*—Traditional large-scale multi-objective optimization algorithms (LSMOEAs) encounter difficulties when dealing with sparse large-scale multi-objective optimization problems (SLMOPs) where most decision variables are zero. As a result, many algorithms use a two-layer encoding approach to optimize binary variable *Mask* and real variable *Dec* separately. Nevertheless, existing optimizers often focus on locating non-zero variable positions to optimize the binary variables *Mask*. However, approximating the sparse distribution of real Pareto optimal solutions does not necessarily mean that the objective function is optimized. In data mining, it is common to mine frequent itemsets appearing together in a dataset to reveal the correlation between data. Inspired by this, we propose a novel two-layer encoding learning swarm optimizer based on frequent itemsets (TELSO) to address these SLMOPs. TELSO mined the frequent terms of multiple particles with better target values to find mask combinations that can obtain better objective values for fast convergence. Experimental results on five real-world problems and eight benchmark sets demonstrate that TELSO outperforms existing state-of-the-art sparse large-scale multi-objective evolutionary algorithms (SLMOEAs) in terms of performance and convergence speed.**

***Index Terms*—Evolutionary algorithms, learning swarm optimization, sparse large-scale optimization, sparse large-scale multi-objective problems, two-layer encoding.**

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I. Introduction

MULTI-OBJECTIVE optimization algorithms (MOEAs) have garnered extensive scholarly attention over numerous years due to their prowess in addressing intricate and nonlinear multi-objective optimization problems (MOPs) [1], [2]. Nevertheless, while MOEAs exhibit promising efficacy in resolving optimization quandaries of modest dimensions [3], [4], their performance experiences significant degradation when confronted with scenarios involving over 100 decision variables. This category of problems, denoted as large-scale multi-objective optimization problems (LSMOPs) [5], is characterized by the exponential augmentation of the search space, precipitated by the escalation of the decision variable's dimension [6]. The inadequacy of effective strategies to navigate this expansive search terrain reduces efficiency to conventional MOEAs [7]. Additionally, the magnification of local optima regions, stemming from the enlargement of the search domain, presents fresh quandaries for conventional MOEAs [8], hastening the onset of premature convergence. As a result, the realm of MOEAs research requires further exploration to effectively confront the problems posed by progressively intricate LSMOPs [9].

Ma *et al.* [10] introduced a novel approach incorporating the categorization of decision variables into the multi-objective optimization algorithms (LSMOEAs). The central tenet involves the separate optimization of each cluster by aggregating variables of congruous attributes. Similarly, WOF [11] undertakes the conversion of the original optimization problem into an optimization weight vector, thus realizing a reduction in the dimensionality of decision variables by assigning specific weights to each cluster. Lü *et al.* [12] proposed the multi-island parallel evolution framework (PEA) with a left-right probability migration based on the idea of decomposition topology to solve large-scale flight allocation problems. Guan *et al.* [13] apply the idea of coevolution to the long-term conflict avoidance of large-scale flight scheduling. SECSO [14] introduced a self-exploratory strategy before the learning phase, enhancing swarm exploration capabilities and circumventing the entrapment of swarms within local optima. Furthermore, Liu *et al.* [15] devised a surrogate-assisted evolutionary algorithm featuring parallel random grouping to resolve the computationally-expensive large-scale feature selection problems.

However, these LSMOEAs are inefficient in solving sparse large-scale multi-objective optimization problems that widely

exist in real life (most variables in Pareto optimal solutions are zero) [16]. Sparse optimization problems are generally characterized by having a much smaller number of non-zero variables (d) in each Pareto optimal solution compared to the total number of decision variables (D), often expressed as $d \ll D$ [17]. This is because LSMOEAs did not consider the characteristics of Pareto optimal solutions in sparse large-scale multi-objective optimization problems (SLMOPs) at the beginning of their design, making it difficult for most LSMOEAs to optimize SLMOPs. Fortunately, the inherent sparsity characteristics of SLMOP typically manifest as prior knowledge before the optimization process. This implies that addressing this challenge can be achieved by designing sparse algorithms. In portfolio optimization [18], investors strive to allocate funds among many available investments strategically. Notably, investment portfolios often exhibit sparse characteristics, as most investors can manage only a limited number of investments effectively. In this scenario, there exists foreknowledge that necessitates the selection of investments with superior cost-effectiveness, leading to the exclusion of the majority of available options. In addition, with a substantial proportion of variables being zero, it markedly diminishes the algorithm's search space, thereby, to some extent, reducing the search complexity of the problem. Unfortunately, effectively locating variables that should be set to non-zero positions has become a new challenge for SLMOPs researchers.

One plausible resolution for addressing the SLMOPs is to embrace a two-layer encoding approach [19]. Tian *et al.* introduced SparseEA [17], an evolutionary algorithm tailored for solving SLMOPs, wherein the decision variables are divided into the real variable *Dec* and binary variable *Mask*. MOEA/PSL [20] proposes MOEAs based on a de-noising automatic encoder (DAE) and restricted Boltzmann machine (RBM) to process SLMOPs. PM-MOEA [21], on the other hand, leverages pattern mining techniques to discern non-zero decision variables. SLMEA [22] utilizes clustering in the decision space to create multiple groups, aiming for dimensionality reduction in ultra-large multi-objective optimization problems. SLMEA incorporates matrix operations designed for improved solutions in sparse settings. S-ECSO [23] integrates strongly convex sparse operators into a competitive swarm optimizer, showing the initial application of a particle swarm optimization algorithm to SLSMOPs.

Meanwhile, Tan and Wang [24] introduce a feature selection operator using non-dominated sorting to select pertinent non-zero decision variables. They employ Kriging-assisted MOEAs to tackle the redefined problem, yielding notably improved performance for sparse large-scale multi-objective optimization algorithms (SLMOEAs), especially with resource-expensive computations. MDR-SAEA [25] unveils a multi-faceted dimensionality reduction strategy encompassing multiple stages. MDR-SAEA conducts independent evaluations for each decision variable, intelligently ascertaining sparsity and identifying non-zero decision variables to reduce problem dimensionality effectually. The SPS algorithm [26], proposed by Kropp *et al.*, demonstrates that initializing a population with optimal or near-optimal sparsity effectively enhances convergence and diversity. However, SPS heavily relies on prior knowledge of objective sparsity parameters, significantly impacting its performance. Subsequently, Kropp *et al.* [27] introduced sampling of sparse populations with varying fringes for population initialization. In this approach, the non-zero variables of the population are evenly distributed in the form of fringes, ensuring that each variable possesses a non-zero value.

Most two-layer encoded SLMOEAs frequently attempt to approximate the sparse distribution inherent in the real Pareto optimal solutions. However, the true aim of optimization lies in optimizing the function value rather than solely ascertaining the sparsity of decision variables. Moreover, the disregard for the interrelation between decision variables results in a focus solely on the sparsity of individual decision variables. This may lead to a degradation of solutions, a drawback particularly pronounced in scenarios characterized by constrained computational resources.

Furthermore, a considerable limitation of most extant SLMOEAs lies in their inefficiency with functional evaluation consumption. This is primarily attributed to elevated expense associated with the functional evaluation of numerous real-world, large-scale sparse multi-objective optimization challenges. Consequently, it is inappropriate for SLMOEAs to use many function evaluations to solve SLMOPs [21].

In data mining, frequent itemsets unveil combinations of items or attributes that frequently co-occur in a dataset [28]. Such associations can shed light on correlations, patterns, or rules within the data. Drawing inspiration from this concept, we introduce a novel approach named the two-layer encoding learning swarm optimizer based on frequent itemsets (TELSO). TELSO mines frequent terms of multiple particles with better objective values to find mask combinations that can obtain better objective values for fast convergence. In summary, this paper makes the following principal contributions.

1) We discussed the limitations of the prevailing two-layer encoding SLMOEAs. Most of these algorithms strive to emulate the sparse distributions akin to real Pareto-optimal solutions but overlook the change in objective function value. Notably, achieving a solution where the binary variable *Mask* closely resembles the real Pareto-optimal solution mask does not inherently guarantee better objective function values compared to an alternative *Mask* with more significant disparities from the real Pareto-optimal solution mask. Persistently approximating the mask composition of the real Pareto-optimal solutions during the optimization process in two-layer encoded SLMOEAs might not yield solutions with superior objective values and could potentially exacerbate the degradation of objective values.

2) We propose an innovative masking learning (ML) strategy based on the frequent itemsets. Generally speaking, outstanding particles indicate a good direction for exploration, and mine their fragments of potentially valuable information hold significance. ML selects a set of outstanding particles for each particle, gathering potentially advantageous information fragments from distinct dimensions. In contrast to conventional methodologies that focus on detection of the sparsity of

individual decision variables to approximate the sparse distribution of the real PS, ML updates the mask with the goal of generating solutions with superior objective values. We then quickly optimize the objective value by learning frequent items of particles with better objective values. Furthermore, the real-time assessment of particle fitness using iterative progression increases the opportunity to learn excellent mask combinations.

3) We propose a dynamic mutation strategy for binary variables suitable for ML. ML adjusts the mutation probability of masks according to different stages of evolution. Initially, frequent items shared by particles have a higher chance of being flipped than the mask with infrequent items. As evolution proceeds, this chance decreases while the opposite chance increases. This helps particles avoid premature convergence due to incorrect mask combinations in the early stages. It also helps particles find optimal mask combinations quickly in later stages.

4) This paper presents a swarm optimizer that uses two-layer encoding learning based on frequent itemsets (TELSO). In each iteration, multiple improved particles are randomly chosen for each particle. First, the binary variable layer is updated by applying an ML strategy to the particle's mask. Then, the real variable layer is updated by learning the best particles to adjust the particle's velocity and position. This ensures that it provides better real value when detecting the position of non-zero variables. Thus, TELSO can simultaneously learn the two-layer mask composition of excellent particles. In other words, the real variables learn and optimize synchronously with the corresponding mask to achieve better objective values.

The remaining parts of this paper are structured as follows. In Section II, we briefly summarize existing PSO-based variants. Additionally, we briefly introduce existing SLMOEAs and discuss their limitations, which is also the motivation for our work. Finally, we introduce the relevant concepts of frequent itemsets. Section III describes the primary procedure and advantages of our proposed TELSO. Section IV reports the experimental results of TELSO and compares it with other SLMOEAs on the benchmark set SMOP and five real-world problems. Finally, in Section V, we summarize the paper and discuss potential avenues for future research.

II. Related Work and Motivation

A. Multi-Objective Optimization Problems

MOPs can be mathematically defined as [29]

$$\begin{cases} \min F(X) = (f_1(X), f_2(X), \ldots, f_m(X)) \\ \text{s.t. } X \in \Omega \end{cases} \tag{1}$$

where $X = (x_1, x_2, \ldots, x_D)$ is a D-dimensional decision variable vector from the decision space Ω. MOPs with $D > 100$ are known as LSMOPs. The objective function vector $F(X)$ comprises m conflicting objectives. The objective is to find a set of trade-off solutions, called the Pareto optimal set (PS), for MOPs. Let X_1, $X_2 \in \Omega$; X_1 is said to dominate X_2, denoted by $X_1 \prec X_2$, if and only if $f_i(X_1) \leq f_i(X_2)$ for each $i \in \{1, \ldots, m\}$ and $f_j(X_1) < f_j(X_2)$ for at least one indicator $j \in \{1, \ldots, m\}$; If all X from Ω cannot dominate X_1, we call X_1 nondominated or PS. The set of all the Pareto optimal objective vectors is called the Pareto optimal front (PF). SLMOPs show that most decision variables in the Pareto optimal solution are zero; The number of non-zero variables d in each Pareto optimal solution is much smaller than the total number of decision variables D ($d \ll D$).

B. PSO-Based Variants

Particle swarm optimization (PSO) [30] has captured researchers' interest owing to its commendable search efficiency and comprehensible operational framework. Subsequently, Zhao *et al.* [31] proposed a new binary particle swarm optimizer to improve the performance of PSO further. While PSO exhibits commendable efficacy in addressing optimization challenges of modest dimensions, its performance diminishes when confronted with dimension-expansive characterized by an excess of 100 decision variables, namely, large-scale optimization problems.

To address this problem, numerous PSO variants have been introduced [32], [33]. Liang *et al.* [33] suggested excluding the *gbest* update strategy in PSO, advocating for only *pbest* to guide particle position updates. Cheng and Jin introduced a one-on-one social learning mechanism into the PSO and conceived SL-PSO [34]. On one facet, this distinctive learning mechanism augments the diversity within the swarm and mitigates the issue of premature convergence encountered in large-scale optimization problems. On another facet, the one-on-one learning mechanism enables the algorithm to explore different regions and generate non-singular Pareto optimal solutions, a virtue advantageous for optimizing MOPs. Following this, in MCSO [35], the tri-competition is employed to update two-thirds of the particles within the population. The update methods for *loser*1 and *loser*2 are identical; both learn from the same winner. Contrastingly, LLSO [36] categorizes particles into groups of equal size based on their fitness. The particles in the lower-level group learn from those in the two higher-level groups. This approach ensures that particles with similar fitness levels do not compete, enhancing convergence efficiency. Ming *et al.* introduced a competitive and cooperative swarm optimizer, enhancing convergence speed through competitive swarms and improving capability to escape local feasible or local optima regions through a cooperative swarm [37]. In TPLSO [38], the learning process is segmented into mass learning and elite learning to handle populations sequentially. Different update methods are applied to the two losers in the tri-competitions during the mass learning stage. Additionally, the winning particles in the competition are further updated in the elite learning phase to obtain an improved winner. The CSOPSO [39] employs non-dominated sorting and crowding distance for elite particle selection. However, it is noteworthy that experiments have been limited to a maximum of 30 decision variables. Tian *et al.* advanced the concept by proposing LMOCSO, based on this one-on-one learning mechanism, to tackle LSMOPs [8]. CPCSO [40] incorporates perturbation and coevolutionary frameworks into competitive groups, ensuring particles maintain a smooth search path. Additionally, S-ECSO is proposed and grounded in the same

learning paradigm, to address SLMOPs [23]. We can see that PSO-based algorithms based on this unique learning mechanism can potentially solve SLMOPs. Therefore, we innovatively applied this learning mechanism based on PSO to mask learning to optimize SLMOPs.

C. Motivation

In the pursuit of maintaining population diversity, the majority of prevailing SLMOEAs embrace a two-layer encoding methodology. This involves utilizing the real vector *Dec* and the binary vector *Mask* to represent each solution x [21]

$$x_i = Dec_i \times Mask_i, \quad i = 1, \ldots, D \tag{2}$$

where the Dec_i represents the real value of the i-th decision variable, while $mask_i$ records the i-th decision variable that should be set to zero. For example, a real vector $Dec = (0.3, 0.2, 0.7, 0.9)$ and a binary vector $Mask = (0, 1, 1, 0)$, the decision variables obtained by combining *dec* and *mask* are $(0, 0.2, 0.7, 0)$. The binary vector *Mask* is used to control the sparsity of the solution [21].

They evaluate a single decision variable's sparsity to approximate the real PS's sparse distribution. However, the objective of the optimization is to minimize (or maximize) the objective function rather than mask composition that is as similar as possible to the sparse distribution of a real PS. Because two-layer encoding is not independent, the objective function value depends on the interaction between binary and real variables. A mask composition as similar to the sparse distribution of real PS as possible does not always mean that the objective function value is better. Therefore, two-layer coded SLMOEAs that constantly approximate the mask composition of real PS during optimization may not quickly optimize the objective value and could even worsen the individual result. Finding the real PS may be challenging, especially when computing resources are limited or real-world problems are complex. Such SLMOEAs may be unable to find solution sets of better objective functions with limited computing resources.

To demonstrate a phenomenon, we consider a sparse minimization problem denoted by $f(x_1, x_2) = |x_1|/2 + (|x_2| - x_1^2)^2$. The first term of the function $|x_1|/2$ can be regarded as a regularization item, which encourages the x_1 value to be zero, thus achieving sparsity. In addition, the second item $(|x_2| - x_1^2)^2$ in the function can also be regarded as a regularization term, which penalizes the difference between x_2 and x_1^2 and makes as close as possible to maintain sparsity. The 3D curved surface of the function is presented in Fig. 1, where the X and Y axes correspond to the decision variables x_1 and x_2, respectively. The Z axis represents the objective function value. The cool and warm color regions signify the small and large objective function values, respectively. Notably, the image in Fig. 1 showcases only the values of x_1 and x_2 ranging from zero to one for convenience. Moreover, in the computation of the objective function value of the function above, we set the nonzero real variable *Dec* to be one. This means that changes in *Dec* are not considered.

As shown in Fig. 1, when the decision variable masks are

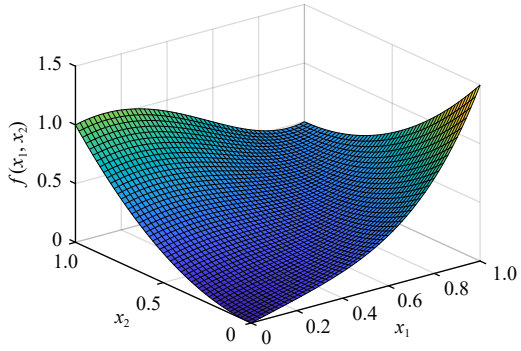


Fig. 1. A minimization problem $f(x_1, x_2) = |x_1|/2 + (|x_2| - x_1^2)^2$ 3D curved surface, where the X and Y axes represent the values of decision variables x_1 and x_2, respectively, and the Z axis represents the objective function value.

(0, 0), (0, 1), (1, 0), and (1, 1), the objective function values are 0, 1, 1.5, and 0.5, respectively. The function reaches its minimum value when the decision variable mask is (0, 0). That is, the *Mask* of the real optimal solution is (0, 0). However, the mask composition (0, 1) and (1, 0) that approximates the sparse distribution of the real optimal solution has a worse objective value than (1, 1), which is entirely different from the *Mask* of the real optimal solution. Therefore, we can make the following inference. A mask combination close to the real PS of mask composition may not mean the objective value is better.

If we consider the large-scale problem, the change in the objective value of the function may be more complex. With augmentation in the count of decision variables, the interrelations among these variables grow intricate, amplifying their impact on the objective function value. We can extend the function to a large-scale example, such as $f(x_1, x_2) = |x_1|/2 + (|x_2| - x_1^2)^2 + |x_3| + \cdots + |x_{d-1}| + |x_d|$. It can be seen that the conclusion is the same as the original conclusion.

It is worth noting that we did not consider the situation where real variable *Dec* changes in the above example. If we consider the variation of *Dec* or a problem which is multiobjective, the function's objective value change may become more intricate. Fig. 2 presents the dominance ranking of particles after the initial iteration of SLMOEAs in optimizing the SMOP6 with 10 decision variables. For ease of reference, the annotation beneath each particle denotes the number of particles it dominates. The decision variables for particles p_1 and p_2 are correspondingly (0.0433, 1.8509, 0, ..., 0) and (0.0433, 0, 0, ..., 0). Their objective values are (0.0025, 0.9990) and (0.0026, 1.0456). From Fig. 2, it can be seen that p_1 has a better dominance ranking than p_2. However, the real Pareto optimal solutions feature a sparse distribution of (1, 0, ..., 0). We can observe that p_2, which aligns more closely with the sparse distribution of the authentic Pareto optimal solutions, possesses a lower dominance ranking. Intriguingly, p_2 is even dominant with particle p_1, which deviates from the sparse distribution characterizing the real Pareto optimal solutions.

Therefore, blindly pursuing a sparse distribution approximating the real PS may be inefficient for optimizing the objective function value. A possible solution is to mine the *Dec* and *Mask* components of excellent particles and simultaneously

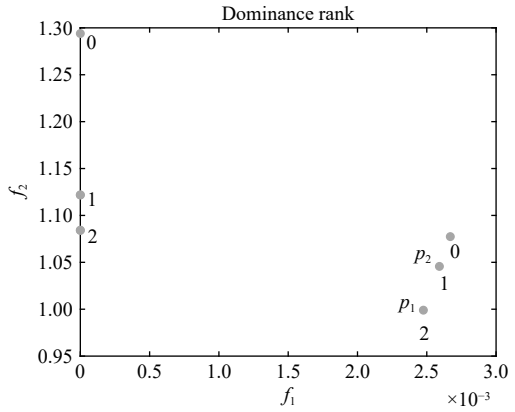


Fig. 2. Comparison of particle dominance rank for the SMOP6 problem with two objectives.

learn their two-layer encoding to obtain better objective function values. This is due to the fact that their binary and real variable often means better exploration directions and objective function values.

D. Frequent Itemset

Frequent itemsets frequently emerge in data mining as a means to reveal correlations within data [41]. A frequent itemset is a collection of frequently co-occurring items in a dataset. It involves an itemset consisting of n items, $I = i_1, i_2, \ldots, i_n$, and a dataset D containing multiple transactions, each a collection of items. All transactions within D are represented by $T = t_1, t_2, \ldots, t_m$, where each transaction t_J is an itemset. The proportion of transactions that contain all the items in a given itemset X is represented by $supp(X)$, which denotes the support level of X. The itemset X is considered frequent if its support is greater than or equal to a predefined threshold of θ, i.e., $supp(X) \geq \theta$. Therefore, for dataset D, an itemset X is a frequent itemset if and only if the number of occurrences of X in D is not less than the predefined threshold of θ. The definition of frequent itemsets can be further expressed as

$$\frac{|\{t_j \in D \mid X \subseteq t_j\}|}{|D|} \geq \theta \tag{3}$$

where $X \subseteq t_j$ represents the number of occurrences of itemset X in each transaction t_j, and $|D|$ represents the total number of transactions in dataset D.

We provide an illustrative example to increase the comprehension of the concept of frequent itemsets. Consider a supermarket that offers a range of items, including bread, milk, eggs, noodles, and coffee. Our goal is to find the support for purchasing two goods: bread and milk. First, we need to scrutinize the transactional data to identify how often customers have purchased bread and milk concurrently. The hypothetical record table is shown in Table I. From these records, we can see three times when customers purchased bread and milk simultaneously (baskets 1, 3, and 6). Therefore, the support purchasing both bread and milk is 3/6, i.e., 0.3 or 30%. In essence, the support measure reflects the frequency of an itemset across all transactions. If an itemset frequently appears in the transactions, it has a high support value; conversely, if it appears infrequently, it has a low support value. If the predefined threshold is set to 30% or less, we consider the itemset Bread, Milk as a frequent itemset.

TABLE I
SUPERMARKET SALES RECORDS

Baskets	Goods
1	Bread, milk
2	Bread, egg
3	Bread, milk, noodles
4	Milk, noodles
5	Bread, egg, coffee
6	Bread, milk, egg, noodles

In data mining, frequent itemsets are often used to find relevant patterns in data, such as market basket analysis and network traffic analysis [42]. By mining frequent itemsets in data, common combinations can be identified for prediction and recommendation. In addition, frequent itemsets can also be used for tasks such as feature selection [43] and classification [44].

Similar its definition in data mining, in this paper, a set of particles that share the same value on a particular dimension is deemed frequent on that dimension, i.e., $\theta = 1$. A vector of multiple frequent dimensions is referred to as a frequent item, and a set comprising multiple frequent items is recognized as a frequent itemset. It is worth noting that the mask learning strategy based on frequent itemsets proposed in this optimizer does not mine the entire swarm' frequent itemsets but mines multiple subswarms. Because the sparsity of decision variables is unknown, mining and learning the frequent itemsets of the entire swarm may make it easier for the swarm to fall into the local optima. In contrast, mining multiple intersecting subswarms is less greedy. It can better balance the exploration and exploitation of the swarm.

III. PROPOSED TELSO

As described above, blindly approximating *Mask* of real Pareto optimal solutions may not be a practical approach due to the mutual dependency of decision variables. In addition, many real-world function evaluations, such as sparse inverse covariance matrix estimation problems [45], and sparse linear regression problems [46], are expensive. Finding the *Mask* that resembles the real PS mask with limited computing resources or using complex real-world problems to optimize the objective function value may be inefficient. Therefore, using as few computing resources as possible is important in finding the optimal solution set. In response to these challenges, we introduce TELSO, which mines frequent itemsets of *Mask* that may be distributed across multiple samples. Furthermore, we achieve fast convergence by mining frequent itemsets with better objective values.

To investigate the potential mechanism of TELSO, a series of particle-related definitions were initially presented. Assuming that S denotes the swarm in TELSO, the *Fitness* compute

equation employed in the optimization process is given as follows [47]:

$$Fitness(i) = \min_{\mathbf{j}\in S\setminus\{\mathbf{i}\}} \sqrt{\sum_{m=1}^{M} (\max\{0, f_m(j) - f_m(i)\})^2} \quad (4)$$

where the current equation denotes $f_m(i)$ as the m-th objective value of particle i, while M represents the maximum number of objectives. As illustrated in (4), a higher value of $Fitness$ for particle i implies greater competitiveness. In addition, the proposed TELSO uses a real vector Dec and a binary vector $Mask$ to represent each solution x.

Algorithm 1 Framework of TELSO

Input: $|S|$ (swarm size), $F_{\max}$ (maximum function evaluations).
Output: S'' (final swarm).
1 $S \leftarrow$ InitializationSwarm($|S|$);
2 $V \leftarrow$ InitializationReferenceVector($|S|$);
3 **while** *termination condition not fulfilled* **do**
4 $S' \leftarrow$ Twolayer Coding Learning(S) (Algorithm 2);
5 $S'' \leftarrow$ Twolayer Coding Variation(S') (Algorithm 3);
6 $S''' \leftarrow$ EnvironmentalSelection(S'', V) (RVEA [48]);
7 **return** S''';

Algorithm 1 presents a comprehensive and concise framework for TELSO. Before implementing TELSO, initialization parameters such as the swarm size ($|S|$) and the maximum number of function evaluations ($F_{\max}$) must be established. Next, the swarm is initialized based on a predetermined swarm size and generates a uniformly distributed reference vector (V) (lines 1 and 2). It should be noted that if the decision variables are encoded as real codes, D solutions are randomly and uniformly generated. In contrast, if the decision variables are encoded as binary, a $D*D$ matrix is generated. An identity matrix of size $D*D$ is also generated for the $Mask$ component. Next, the two-layer encoding learning strategy based on frequent itemsets in Algorithm 2 and the two-layer encoding mutation strategy in Algorithm 3 are sequentially executed (lines 3−5) until the termination condition is satisfied. Finally, environmental selection (line 6) is carried out, and the swarm obtained after the last environmental selection is output as the final result. The fundamental steps of the TELSO can be summarized as follows:

1) In the first step of the optimization process, all particles are evaluated for their respective fitness using (4) and sorted in ascending order. The best-performing particles' resulting sequence is s, while the sequence of the worst-performing particles is denoted as one. Following this, for each particle, p_i with a designated sequence number i (initialized as one), two particles with sequence numbers bigger than p_i are randomly selected and labeled as p_{o1} and p_{o2} (with $o1 \neq o2$). It was required that the fitness value of p_{o1} and p_{o2}, represented by $f(p_{o1})$ and $f(p_{o2})$, respectively, satisfy the condition $f(p_i) \leq f(p_{o1})$ and $f(p_i) \leq f(p_{o2})$. Notably, particles with sequences s and s–1 lack two particles with higher sequence numbers. Consequently, they will remain unaltered during the iterative process. From the swarm's perspective, this approach is also justified. The two particles with the highest sequence numbers represent the top two solutions in the swarm, and retaining them helps maintain the superiority of the swarm.

2) Subsequently, for the binary variable $Mask$, particle i learns the frequent items of $p_{o1}.mask$ and $p_{o2}.mask$. This is achieved by mining the frequent item between $p_{o1}.mask$ and $p_{o2}.mask$. A uniformly distributed random number between zero and one is then generated, and if it is less than 0.5, the parts of p_{o1} and p_{o2} masks for the frequent item are assigned to the corresponding dimension of the $Mask$ in the decision variable of particle i. Conversely, if the random number is greater than or equal to 0.5, the parts of the p_{o1} and p_{o2} masks with a value of zero for the frequent item are assigned to the corresponding dimension of the mask in the decision variable of particle i. Particle i is then used according to the following equation to update the real variable Dec:

$$\begin{aligned} V_i(t+1) &= r_0(t)V_i(t) + r_1(t)(P_{o1}(t) - P_i(t)) \\ &\quad + r_2(t)(P_{o2}(t) - P_i(t)) \\ P_i(t+1) &= P_i(t) + V_i(t+1) \end{aligned} \quad (5)$$

where V_i and P_i denote the velocity and position of the particle selected with sequence i. r_0, r_1, and r_2 are uniform random values generated from the interval [0, 1]. P_{o1} and P_{o2} are two outstanding particles randomly chosen from the swarm whose fitness values are significantly better than that of P_i. It should be noted that the P_{o1} and P_{o2} sequences are distinct. The variable t represents the current iteration number. Each particle uses (5) to adjust its velocity and position to approach the optimal solutions.

3) Afterward, a mutation strategy is employed for the particle i. Precisely, the current iteration counts t (t is the ratio of the current number of function evaluations to the maximum number of function evaluations) which is contrasted with a randomly generated number that conforms to a uniform distribution within the interval [0, 1]. Suppose t is less than the aforementioned random number, where one dimension of the frequent items in particle $p_i.mask$ is randomly selected, and its state is flipped. Conversely, if t is greater than or equal to the randomly generated number, one of the infrequent items within $p_i.mask$ will be flipped. It should be emphasized that infrequent items may not exist during the later stages of swarm evolution, implying that no masks in the swarm will undergo a reversal.

4) Finally, if i is less than s–2, then i+1, and Steps 1)–3) will be repeated until the first s–2 particles have completed their learning process and entered the environmental selection stage with the original swarm.

In Step 1), we selected two outstanding particles for each particle p_i. Generally, outstanding particles indicate better objective function values and are a promising direction for exploration [49], as their potentially useful information fragments are more valuable. This is due to the fact that the $Mask$ of outstanding particles usually can generate a better objective value. Randomly selecting prominent particles for each particle and mining their potentially beneficial evolutionary information is more conducive to objective value optimization. In addition, Since different samples may have different segments of potentially useful information, learning the same

excellent components from multiple superior particle genes can be more advantageous, further ensuring the excellence of the mask composition being learned. Lastly, learning from two different particles is more conducive to maintaining swarm diversity in the real variable Dec.

In Step 2), ML differs from conventional practice in two-layer encoding algorithms, in which judgment is based on the sparsity of a single decision variable. Existing SLMOEAs usually assess single decision variables' sparsity and flip masks. This approach entails changing a single mask during the mask update process to find a mask composition approximating real sparsity. In addition, they typically update masks to approximate the mask composition of real Pareto optimal solutions. However, decision variables are interdependent, and traditional two-layer encoding algorithms might not capture their interactions. In other words, blindly flipping an individual's $Mask$ to match the mask of the real PS may result in worse objective function values. The above issues have been discussed in detail in Section II-C.

This paper introduces a novel approach, ML, to address the abovementioned issues. In ML, we update the set of decision variables by mining the frequent terms of multiple prominent particles rather than flipping a single decision variable. Because the frequent term of multiple prominent particles may often be a critical factor in reducing the objective value, it often indicates the evolutionary direction of better objective functional values. Furthermore, mining frequent terms in the $Mask$ of multiple outstanding particles can help capture helpful mask components in different samples, thereby increasing the likelihood of retaining potentially helpful information in various samples.

Algorithm 2 ML

Input: S (current swarm).
Output: S' (new swram).
1 $Fitness \leftarrow$ Calculate fitness according to (4);
2 $S \leftarrow$ Sort by $Fitness$;
3 **for** i : 1 *to* $(|S|-2)$ **do**
4 $p_{o1}, p_{o2} \leftarrow$ Randomly select two particles with fitness better than p_i;
5 /*Mining frequent itemsets*/
6 $Fi \leftarrow$ Mining indexes of variables where $p_{o1}.mask$ and $p_{o2}.mask$ are both zero or non-zero in the same dimension;
7 /*Update $Mask$*/
8 **if** *rand* < *0.5* **then**
9 Update the $Mask$ of the particle p_i corresponding to the position is one of the Fi;
10 **else**
11 Update the $Mask$ of the particle p_i corresponding to the position is zero of the Fi;
12 /*Update Dec*/
13 $p_i.Dec \leftarrow$ Update the velocity and position of particle p_i according to (5);
14 $p_i \leftarrow$ Generate a new particle by multiplying $p_i.dec$ by $p_i.mask$;
15 $S' \leftarrow S' \cup p_i$;
16 **return** S';

Although TELSO is conceptually straightforward, to provide a more elaborate description of the TELSO algorithm, we reveal the decision variable update process in ML in Fig. 3. Within this figure, p_i represents the selected particle, while p_{o1} and p_{o2} denote two particles boasting superior fitness compared to p_i. Their respective masks are illustrated as (0, 1, 0, 1, 1, 0, 1, 0, 1, 0, 1, 0, 1) and (1, 1, 0, 0, 1, 1, 1, 0, 0). We start by extracting frequent items from the samples p_{o1} and p_{o2}, which yield $x_2 = 1, x_3 = 0, x_6 = 1, x_8 = 1$, and $x_9 = 0$ (depicted in dark gray). Subsequently, we allocate a binary coding mask to the corresponding dimensions of $p_i.mask$, wherein x_2, x_3, x_6, x_9, and x_9 of $p_i.mask$ are assigned values of (1, 0, 1, 1, 0) respectively. Notably, the velocity and position of $p_i.dec$ will update based on $p_{o1}.dec$ and $p_{o2}.dec$, as indicated in (5). This ensures the update of the binary variable $Mask$ while simultaneously optimizing the real variable Dec. This is consistent with the starting point of the SparseEA2 design and further demonstrates the soundness of our ML [50]. Algorithm 2 describes the detailed process of ML.

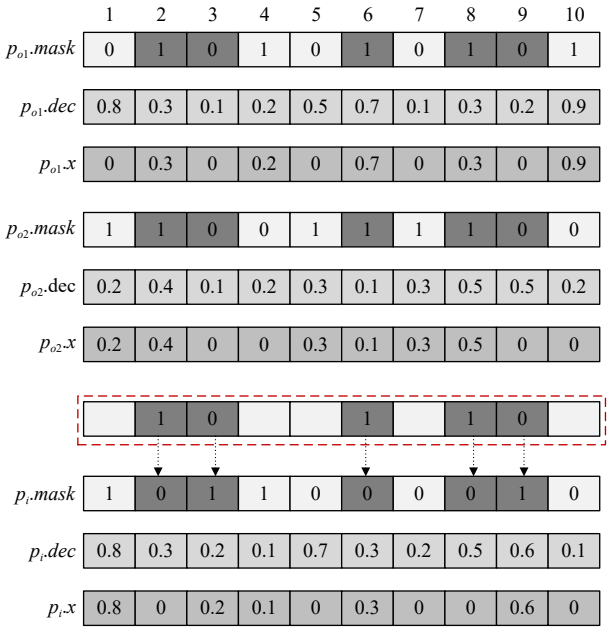


Fig. 3. Particle p_i learns from two better particles p_{o1} and p_{o2} according to the two-layer encoding learning strategy based on frequent itemset, in which the frequent item in $p_{o1}.mask$ and $p_{o2}.mask$ are displayed with the dark gray background.

In Step 3), we propose tailored mutation strategies for ML strategy. The key objective of the mutation strategy is to alleviate the risk of the algorithm converging to suboptimal solutions. However, in most SLMOEAs, binary mask mutation strategies are designed to accelerate swarm convergence. Therefore, their mutation strategy may not adequately facilitate swarm diversification, nor can it help the swarm jump out of the local optimal region when the sparsity evaluation of decision variables is inaccurate.

We propose a mutation strategy, termed MM, to enhance the sparsity of solutions in a swarm by adaptively adjusting the binary variable $Mask$ at different stages of evolution. Simply put, the flipped term depends on the relative size of the

random number and t. In the early stages of population evolution, the t is relatively small. Therefore, there is a high probability that t is less than a random number between 0 and 1. At this point, a random variable in the frequent items is flipped. As evolution progresses, the t gradually increases. Therefore, t only has a small probability of being less than a random number between 0 and 1. At this point, a random variable in the infrequent items is flipped.

At the initial stage, the mutation probability of frequent item flipping of particles is assigned a higher value than infrequent item flipping. As the evolution progresses, the mutation probability of the frequent item flipping of particles gradually decreases. In contrast, the probability of flipping the infrequent item increases proportionally. This adaptive variation strategy caters to the specific needs of each evolutionary stage. It helps prevent particles from maturing prematurely due to improper learning in the early stages. Moreover, the MM helps particles correct the wrong mask in the late stages of swarm evolution. Algorithm 3 describes the detailed process of MM.

Algorithm 3 MM

Input: S' (current swarm), $iter$ (current iterations).
Output: S'' (new swram).
1 **for** i : 1 *to* $|S'|$ **do**
2 **if** $iter < rand()$ **then**
3 $j \leftarrow$ Select an element at random in F_i;
4 **else**
5 $\overline{F_i} \leftarrow$ Find out the set of elements in D except for the F_i;
6 $j \leftarrow$ Select an element at random in $\overline{F_i}$;
7 $p_i.mask \leftarrow$ Flip the j-th binary vector of the i-th particle in the $Mask$;
8 $S'' \leftarrow S'' \cup p_i$;
9 **return** S'';

In Step 4), the two-layer encoding of all particles, excluding the two best-performing ones, undergo updating via ML. It is essential to observe that the particles in the s–1 and s sequences cannot learn frequent itemsets of other particles, given that they do not encounter more outstanding particles. All such particles undergo mutation and enter environmental selection. It is worth noting that the proposed TELSO can also use excellent environment selection strategies other than RVEA. As environment selection is not a core part of TELSO, interested readers can read [48] for more algorithm details.

IV. Empirical Studies

In this section, we assess the efficacy and computational efficiency of the proposed TELSO and four classical SLMOEAs using the widely employed sparse large-scale benchmark SMOP [17] and five distinct real-world problems. Notably, SMOP is the pioneer benchmark designed explicitly for evaluating SLMOEAs and serves as testing for algorithmic performance in SLMOEAs such as MOEA/PSL and S-ECSO.

We shall delineate the experiment into two distinct parts. In the first part, we shall introduce the experimental setup on the benchmark set SMOP1-SMOP8. We selected four advanced and different types of SLMOEAs, in comparison with TELSO, and provided a brief introduction for each of the SLMOEAs. Parameter settings for the experiment have been expatriated in Section IV-A, as well as the employed performance index. Subsequently, in Section IV-B, we shall compare the proposed TELSO and the four most advanced SLMOEAs' overall performance. We also tested the runtime and computational efficiency of five SLMOEAs and presented and discussed the experimental results in Sections IV-C and IV-D. The ablation experiments conducted on the entire algorithm to measure the effectiveness of different TELSO subcomponents are also presented and discussed in Section IV-E.

In the second part, we aim to evaluate TELSO's performance in solving real-world problems. To this end, we conducted experiments on five real-world problems of up to 5120 dimensions using five sparse large-scale algorithms.

A. Experiments Settings on the SMOP1–SMOP8

We selected classic MOEAs in the field of sparse large-scale optimization, which represents four different ideas for solving SLMOPs. First, among the selected MOEAs, SparseEA [17] differentiates decision variables into two-layers and optimizes real and binary variables individually through decision variable scoring. It is worth noting that SparseEA and TELSO use the same initialization strategy. The comparison with SparseEA can better reflect the superiority of the proposed TELSO performance. Second, MOEA/PSL [20] approaches the sparse conundrum by assimilating knowledge from the Pareto optimal subspace. Third, PM-MOEA [21] utilizes pattern-mining evolutionary tactics to mine nonzero decision variables in the present population. Finally, S-ECSO [23] is a sparse variant grounded in CSO, which operates through particle updates rooted in learning. Moreover, S-ECSO shares similarities with the proposed TPLSO, which also employs particle learning. We, through comparison with S-ECSO, demonstrate the effectiveness of TPLSO.

In the interest of impartiality, we ensured that all algorithms under consideration in our experiment employ the source code proffered by PlatEMO [51]. Moreover, all experiments were conducted in the identical version of PlatEMO. Notably, our TELSO and the four other SLMOEAs were bereft of adjustable parameters. This feature diminishes the parameter selection burden during experimentation while mitigating the influence of our experimental outcomes being subject to adjustment parameters to some extent.

To provide additional evidence of the scalability of TELSO, we conducted a series of experiments using SMOP benchmark sets in various dimensions. First, we set θ to zero and used a swarm size 100 for all experiments. Next, we generated functions with decision variable dimensions 500, 1000, and 5000, where the corresponding function evaluation settings are $10*D$ to assess the algorithm's convergence rate with swarm distribution under limited computational resources. Our experimental results demonstrate that TELSO exhibits a compelling convergence rate and diversity preservation in all tested scenarios.

This study employs the widely-accepted performance indi-

TABLE II
IGD VALUES OBTAINED BY MOEA/PSL, PM-MOEA, S-ECSO, SPARSEEA, AND TELSO ON TWO OBJECTIVES SMOP1–SMOP8 WITH 500, 1000, AND 5000 DECISION VARIABLES, WHERE THE BEST RESULT IN EACH ROW IS HIGHLIGHTED. "+", "−", AND "≈" INDICATE THAT THE RESULT BY ANOTHER SLMOEA IS BETTER THAN, WORSE THAN, OR COMPARABLE TO THAT OBTAINED BY TELSO

Problem	M	D	MOEA/PSL	PM-MOEA	S-ECSO	SparseEA	TELSO
SMOP1	2	500	1.4206e−2 (3.47e−2)−	4.1284e−3 (1.72e−4)−	4.5941e−3 (2.07e−4)−	5.0904e−3 (1.85e−3)−	3.7361e−3 (1.58e−4)
		1000	2.7044e−2 (4.25e−2)−	4.0521e−3 (1.63e−4)−	3.9082e−3 (3.25e−4)≈	4.7107e−3 (1.85e−4)−	3.8384e−3 (2.22e−4)
		5000	4.7592e−3 (3.34e−5)−	3.8420e−3 (1.97e−5)≈	3.8964e−3 (6.18e−5)≈	4.6002e−3 (5.19e−5)−	3.7906e−3 (5.15e−4)
SMOP2	2	500	7.5053e−3 (8.78e−3)−	4.1217e−3 (8.62e−5)−	4.7426e−3 (1.29e−4)−	5.0845e−3 (7.15e−4)−	3.5899e−3 (2.07e−5)
		1000	9.3824e−3 (1.15e−2)−	4.0272e−3 (1.25e−4)−	3.9380e−3 (2.44e−4)−	4.9785e−3 (7.45e−4)−	3.6977e−3 (2.09e−4)
		5000	1.8845e−1 (2.59e−1)−	3.7354e−3 (3.14e−5)−	3.8924e−3 (5.09e−5)−	4.4043e−3 (1.35e−4)−	3.6915e−3 (5.75e−5)
SMOP3	2	500	3.1583e−2 (8.94e−2)−	1.2117e−1 (2.38e−1)−	4.4502e−3 (1.17e−4)−	2.5759e−2 (5.16e−3)−	3.5765e−3 (9.87e−6)
		1000	6.6020e−2 (2.24e−1)−	1.6343e−2 (4.01e−2)−	1.6758e−1 (2.37e−1)−	1.5886e−2 (4.75e−3)−	3.5758e−3 (3.37e−6)
		5000	4.6292e−3 (1.27e−5)−	3.7173e−3 (6.81e−5)−	3.7483e−3 (1.27e−4)−	4.2190e−3 (1.30e−4)−	3.5747e−3 (7.14e−6)
SMOP4	2	500	6.0775e−3 (1.87e−3)−	4.4304e−3 (1.67e−4)−	4.8474e−3 (2.07e−4)−	5.1176e−3 (7.18e−4)−	3.9326e−3 (1.26e−5)
		1000	9.3496e−3 (1.58e−2)−	4.4507e−3 (1.46e−4)−	4.0674e−3 (1.72e−4)≈	4.9960e−3 (2.63e−4)−	4.0489e−3 (2.68e−4)
		5000	5.4516e−3 (1.55e−4)−	4.2096e−3 (1.62e−4)−	4.1444e−3 (2.41e−4)−	4.8876e−3 (3.74e−4)−	3.9745e−3 (3.48e−5)
SMOP5	2	500	1.7149e−2 (2.06e−2)−	4.5517e−3 (2.06e−4)−	4.8300e−3 (2.84e−4)−	5.0475e−3 (4.08e−4)−	3.9332e−3 (1.25e−5)
		1000	1.4781e−2 (2.09e−2)−	4.3986e−3 (1.27e−4)−	4.1079e−3 (1.30e−4)−	4.8361e−3 (1.92e−4)−	3.9563e−3 (4.87e−5)
		5000	5.2282e−3 (5.10e−4)−	4.3241e−3 (5.43e−5)−	4.0112e−3 (3.21e−5)−	4.6700e−3 (1.70e−4)−	3.9154e−3 (3.47e−5)
SMOP6	2	500	6.0494e−3 (1.89e−3)−	4.5503e−3 (2.30e−4)−	4.7470e−3 (2.52e−4)−	4.8978e−3 (1.98e−4)−	3.9353e−3 (1.94e−5)
		1000	7.2140e−3 (5.82e−3)−	4.3449e−3 (1.41e−4)−	4.0692e−3 (1.07e−4)−	4.8684e−3 (2.47e−4)−	3.9794e−3 (6.47e−5)
		5000	4.9314e−3 (2.18e−4)−	4.1161e−3 (7.80e−6)−	3.9886e−3 (1.04e−4)≈	4.6815e−3 (1.14e−4)−	3.9615e−3 (4.48e−5)
SMOP7	2	500	9.8182e−2 (1.80e−1)−	1.2864e−2 (1.32e−2)−	5.1345e−3 (2.57e−4)−	5.9920e−3 (1.24e−3)−	4.2007e−3 (2.67e−4)
		1000	1.0326e−1 (1.58e−1)−	1.3503e−2 (3.49e−2)−	6.8949e−3 (5.70e−3)−	5.5663e−3 (6.15e−4)−	4.6085e−3 (9.80e−4)
		5000	1.2126e−1 (1.64e−1)−	4.7424e−3 (1.16e−4)−	4.2465e−3 (5.57e−5)≈	5.0591e−3 (5.59e−5)−	4.1476e−3 (6.76e−4)
SMOP8	2	500	2.7423e−2 (3.32e−2)−	1.7694e−2 (1.84e−2)−	5.1453e−3 (4.75e−4)−	9.5668e−3 (2.63e−3)−	4.0019e−3 (6.04e−5)
		1000	3.1970e−2 (8.60e−2)−	6.5453e−3 (3.46e−3)−	1.2746e−1 (1.55e−1)−	7.4817e−3 (2.10e−3)−	4.0697e−3 (1.86e−4)
		5000	5.4746e−3 (1.11e−5)−	4.1820e−3 (8.37e−5)−	4.5673e−3 (5.01e−4)−	4.9721e−3 (3.63e−4)−	4.0414e−3 (1.46e−5)
+/−/≈			0/24/0	0/23/1	0/19/5	0/24/0	

cator, the inverted generational distance (IGD) [52], to assess convergence and uniformity performance. To compute IGD, approximately 10 000 reference points are sampled on each PF of SMOP [53]. Moreover, we analyzed the experimental results statistically using the Wilcoxon rank-sum [54] test with a significance level of 0.05. In this analysis, the symbols "+", "−", and "≈", respectively indicate that the compared algorithm's performance is significantly better than, significantly worse than, or similar to that of the algorithm proposed in our study.

B. Performance Comparison With SLMOEAs

The PF shapes of eight benchmark sets SMOP are distinct. Specifically, the PF of SMOP1–SMOP3 is linear, the PF of SMOP4–SMOP6 is convex, and the PF of SMOP7 and SMOP8 is concave. To thoroughly evaluate the proposed TELSO's performance, we independently executed TELSO and four of the most advanced SLMOEAs: MOEA/PSL, PM-MOEA, S-ECSO, and SparseEA, on the sparse large-scale benchmark suite SMOP1–SMOP8 for 30 runs. Table II presents the experimental results for two objectives, highlighting the best results for each test case in gray. Based on the results in Table II, we derive the following observations for the eight SMOP components.

1) Table II reveals that the proposed TELSO outperforms the other most advanced SLMOEAs in all two-objective SMOP problems and achieves optimal solutions (highlighted in gray) across all cases. In contrast, the other four algorithms failed to obtain optimal values. This is due to the frequent items mined by TELSO, which can help the swarm quickly find better mask combinations of objective function values and guide the evolution direction of the swarm. Thus, we can conclude that TELSO is competitive with the existing classical SLMOEAs regarding convergence and distribution. In addition, it is worth noting that TELSO performs well on different PF shapes of SMOP, including the multi-modal SMOP1, SMOP2, SMOP4, and SMOP6, as well as the uni-modal SMOP5, demonstrating its capability in solving different types of optimization problems.

2) Based on the results presented in Table II, we observe that TELSO outperforms MOEA/PSL, PM-MOEA, and SparseEA in all 24 problems, as demonstrated by the Wilcoxon rank-sum test. Specifically, compared to MOEA/PSL, which relies on Pareto optimal subspace learning, PM-MOEA, which employs pattern mining technology, and SparseEA, which utilizes a two-layer encoding optimization model simi-

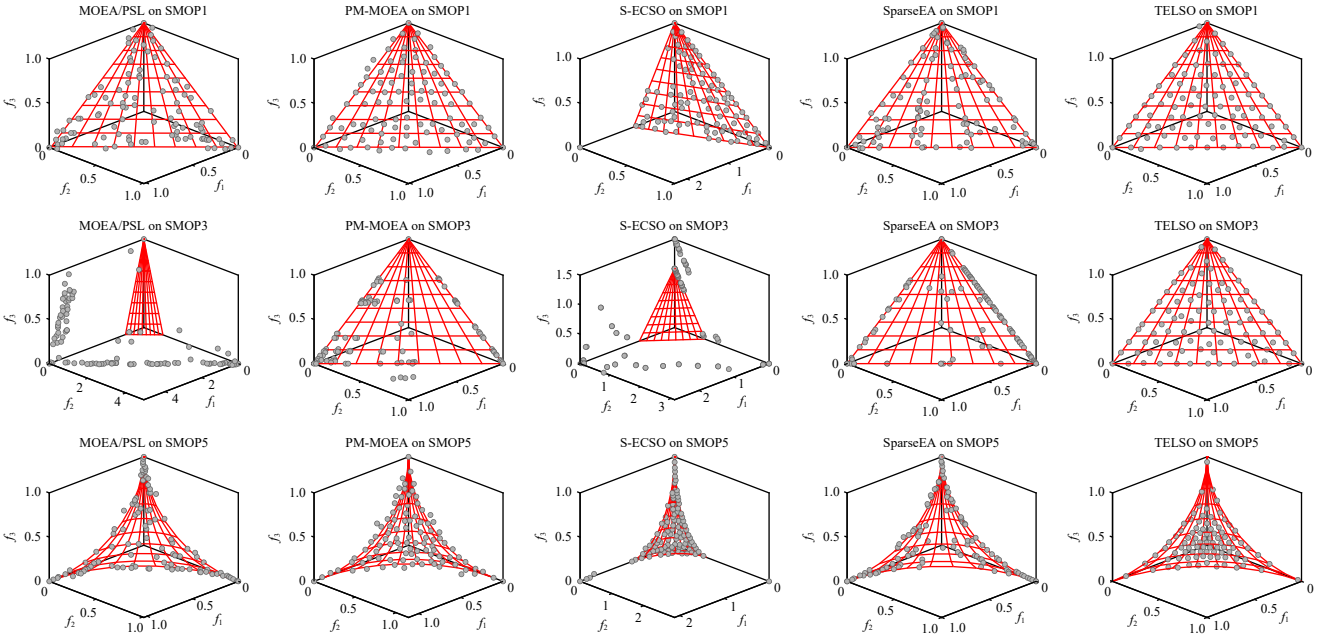


Fig. 4. Nondominated solutions with the median IGD values obtained by MOEA/PSL, PM-MOEA, S-ECSO, SparseEA, and TELSO on three objective SMOP1, SMOP3 and SMOP5 with 1000 decision variables.

lar to TELSO, TELSO yields superior performance. Furthermore, compared to S-ECSO, another PSO-based variant, TELSO outperforms it in 19 out of 24 problems and performs similarly in SMOP1 and SMOP4 with 1000 decision variables. TELSO performs better than the four classic SLMOEAs using a simple and comprehensible optimization method.

In order to further demonstrate the efficacy of TELSO in tackling more complex three objectives SMOP, we present the non-dominated solutions in Fig. 4 obtained by optimizing the SMOP benchmark set with three objectives and 1000 decision variables using five algorithms. Furthermore, to better showcase the quality of the solutions obtained by the algorithms, we highlight the true PF of the problem with red lines or planes composed of lines. The following observations about SMOP can be drawn from the results presented in Fig. 4.

1) As seen from Fig. 4, MOEA/PSL and S-ECSO failed to achieve convergence when optimizing the three objective SMOP3 problem. Despite PM-MOEA and SparseEA achieving convergence, Pareto solutions are mainly distributed on the edge of the true PF. In contrast, the proposed TELSO performs well in the three objective SMOP3 problems and ensures good solution distribution. This indicates that the mask combination learning mechanism adopted by TELSO can accelerate the convergence of the swarm.

2) Fig. 4 shows the first and third lines, when optimizing SMOP1 and SMOP5, with three objectives. Notably, these algorithms exhibit significantly different convergence behaviors. For instance, S-ECSO, a PSO variant, fails to converge on both problems. When solving SMOP5 with a convex PF, MOEA/PSL and SparseEA yield solutions concentrated in the edge region. At the same time, PM-MOEA based on pattern mining technology does not converge entirely on f_1 and f_2. In contrast, the proposed TELSO outperforms the other four algorithms with the number and distribution of solutions.

C. Runtime Efficiency Comparison With SLMOEAs

Furthermore, we assessed runtime efficiency for five distinct SLMOEAs, as depicted in Fig. 5. To facilitate a more perspicuous presentation of the test outcomes, we employed the Friedman test [55] to appraise the runtime of the five algorithms. Succinctly stated, the proposed TELSO exhibits superior runtime efficiency compared to MOEA/PSL, PM-MOEA, and SparseEA. Notably, the two SLMOEAs grounded in PSO exhibit less runtime consumption compared to the remaining three SLMOEAs reliant on evolutionary algorithms. It is plausible to infer that PSO-based algorithms show heightened efficiency with runtime in the optimization of SLMOPs. While S-ECSO surpasses the proposed TELSO in runtime, TELSO significantly outshines S-ECSO in terms of overall performance.

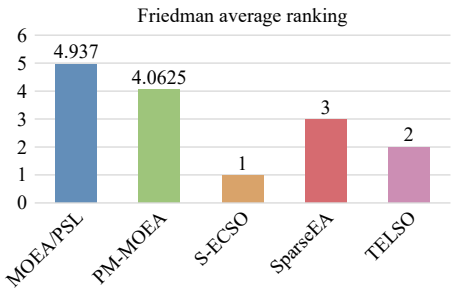


Fig. 5. The average ranking was obtained through Friedman testing of the running time of five SLMOEAs, with a smaller value indicating a shorter running time.

D. Computational Efficiency Comparison With Sparse Large-Scale MOEAs

SLMOEAs suffer from reduced search efficiency due to the exponential growth of high-dimensional space as the number

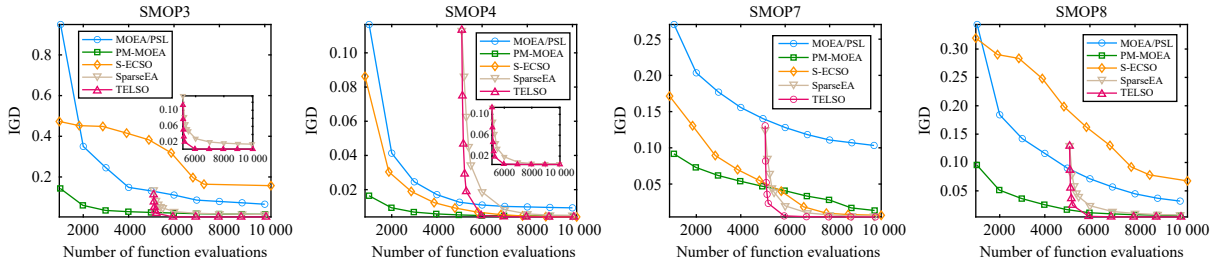


Fig. 6. Convergence profiles of IGD values obtained by five algorithms on two objective SMOP3, SMOP4, SMOP7 and SMOP8 with 1000 decision variables.

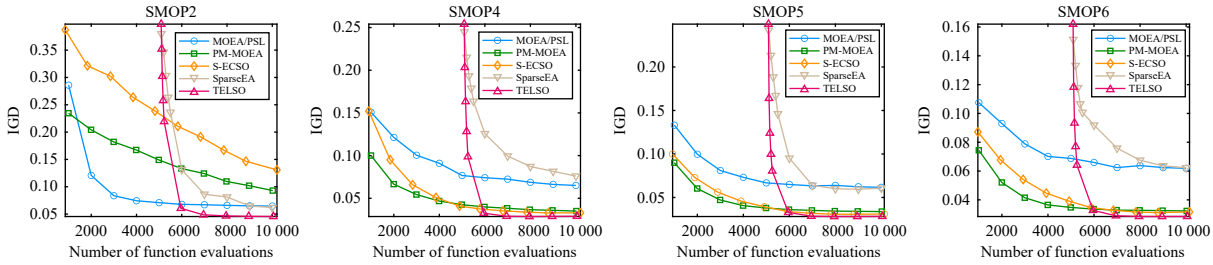


Fig. 7. Convergence profiles of IGD values obtained by five algorithms on three objective SMOP2, SMOP4, SMOP5 and SMOP6 with 1000 decision variables.

of dimensions increases. Moreover, ensuring that the algorithm solves the problem under these constraints in scenarios with limited or expensive computing resources becomes essential. Therefore, optimizing solution quality in scenarios with limited computing resources is critical. Figs. 6 and 7 depict the convergence curves of the IGD values of the five algorithms, demonstrating their efficiency in solving the 1000-dimensional SMOP problem with 10 000 function evaluations. The ensuing discussion follows.

1) Figs. 6 and 7 show the convergence curves of the IGD values of five algorithms with 0−10 000 function evaluations. The results show that the proposed TELSO and SparseEA are inferior to the other three algorithms in the function evaluation of for 0−5000 function evaluations. This is caused by the initialization strategy of the two algorithms. However, for 5000−6000 function evaluations, the IGD value of TELSO rapidly dropped to a level similar to that of other algorithms. This is due to itemsets containing excellent particles, which can help particles quickly find mask combinations with lower objective function values.

2) Fig. 6 indicates that for SMOP3, SMOP7, and SMOP8, the IGD curves of MOEA/PSL and S-ECSO are consistently higher than those of the other algorithms, indicating inferior performance. Although SparseEA shows a rapid decrease in IGD value after initialization, it can be observed that the IGD curve of TELSO remains below that of SparseEA during the evaluation of about 5000–10 000 function evaluations. This may be attributed to the fact that these algorithms cannot efficiently locate the position of non-zero decision variables. Furthermore, as SMOP3, SMOP7, and SMOP8 share a similar problem structure to pattern mining, their landscape functions can only be minimized when specific decision variables are set to zero. Therefore, it can be inferred that TELSO can not only quickly find solutions with better objective values but also effectively locate the positions of non-zero decision variables when solving SMOP problems that are simpler than real-world problems.

3) The convergence curves of IGD values of five algorithms with 0−10 000 function evaluations for SMOP4, SMOP5, and SMOP6 are shown in Fig. 7. SparseEA has a consistently larger IGD value than the other algorithms, indicating that it cannot converge quickly within the limited function evaluation. This may be due to the insufficient searchability of other algorithms. For example, the landscape function of SMOP4–SMOP6 is similar to the feature selection problem, where the sparsity of the solution is directly related to the landscape function. SparseEA cannot solve such problems effectively due to the limited connection between real and binary variables. Conversely, TELSO optimizes real variables as binary variables flip, resulting in a lower IGD value curve, which verifies the effectiveness of our two-level coding learning strategy. Furthermore, the IGD value of TELSO drops rapidly to a lower value after the initial swarm, demonstrating its superior performance compared to the other four algorithms in evaluating approximately 5000−10 000 functions. Therefore, our TELSO can quickly find sparse solutions and search for non-zero variables.

E. Ablation Study Results

We performed an ablation analysis to assess the collective influence of the subcomponents ML and MM on TELSO. This investigation encompasses three distinct configurations: 1) Complete TELSO implementation (TELSO); 2) TELSO without MM (NoMM); and 3) TELSO without ML (NoML). Each configuration gauges the significance of the respective component concerning the algorithm's holistic performance. We executed SMOP1–8 with 1000 decision variables, iterated 30 times each. The ensuing Table III provides the following insights.

1) On the one hand, the ablation study underscores ML's paramount influence within TELSO. Across all test instances,

TABLE III
ILLUSTRATIVE SCENARIO OF THE ABLATION STUDY

Problem	M	D	NoMM	NoML	TELSO
SMOP1	2	1000	3.8940e−3 (5.25e−4)≈	2.2373e−2 (1.77e−2)−	3.8384e−3 (2.22e−4)
SMOP2			3.7866e−3 (5.57e−4)≈	2.7750e−2 (2.50e−2)−	3.6977e−3 (2.09e−4)
SMOP3			3.7269e−3 (3.43e−3)≈	3.5877e−2 (3.55e−2)−	3.5758e−3 (3.37e−6)
SMOP4			5.2150e−3 (2.87e−4)−	1.8519e−2 (2.38e−2)−	4.0489e−3 (2.68e−4)
SMOP5			5.5396e−3 (4.18e−4)−	2.1766e−2 (2.21e−2)−	3.9563e−3 (4.87e−5)
SMOP6			4.9751e−3 (4.62e−4)−	1.8579e−2 (1.07e−2)−	3.9794e−3 (6.47e−5)
SMOP7			5.5746e−3 (1.07e−3)−	3.2394e−2 (2.68e−2)−	4.6085e−3 (9.80e−4)
SMOP8			4.2674e−3 (1.88e−3)≈	4.5179e−2 (4.05e−2)−	4.0697e−3 (1.86e−4)
	+/−/≈		0/4/4	0/8/0	

TABLE IV
THE OBJECTIVE FUNCTION OF THE REAL-WORLD PROBLEMS USED IN THE EXPERIMENT

Problem	Objective function	Parameter meaning	
NN	$Minimize \quad f_1(x) = \frac{\|x\|_0}{D}$ $f_2(x) = ErrorRate(x)$	x: the neural network's weights	$\|x\|_0$: the nonzero weights' number
		D: the total weight's number	$ErrorRate(x)$: the error rate
		f_1: the nonzero weights' ratio	f_2: the error rate on training set
PO	$Minimize \quad f_1(x) = \sum_{i=1}^{D}\sum_{j=1}^{D} x_i\sigma_{i,j}x_j$ $f_2(x) = 1 - \sum_{i=1}^{D} x_i r_i$	x: each instrument's ratio of the portfolio	$\sigma_{i,j}$: the i-th and j-th instruments' covariance
		r_i: the i-th instrument's expected return	f_2: the expected return's negative value
		f_1: the total portfolio's risk	
SR	$Minimize \quad f_1(x) = \frac{\|x\|_0}{D}$ $f_2(x) = \|y - Ax\|_2^2$	x: the reconstructed signal	D: the reconstructed signal's length
		A: the sensing matrix	f_2: the reconstructed signal's loss
		f_1: the reconstructed signal's sparsity	
FS	$Minimize \quad f_1(x) = \frac{\|x\|}{D}$ $f_2(x) = ErrorRate(x)$	x: the selected features	D: the total number of features,
		$ErrorRate$: the error rate of the classifier	f_2: the error rate of the classifier on validation set
		f_1: the ratio of selected features	
KP	$Maximize \quad f(x) = (f_1(x), f_2(x), \ldots, f_n(x))$ $f_i(\boldsymbol{x}) = \sum_{j=1}^{m} p_{i,j} \cdot x_j$	n: the number of knapsacks	m: the number of items
		$p_{i,j}$: profit of item j according to knapsack i	$x_j = 1$: the item j is selected
		$f(x)$: the profit from any number of knapsacks	$f_i(x)$: the profit of knapsack i

removing ML conspicuously enhances the IGD value, signifying a notable decline in TELSO's performance. This experimental result robustly underscores ML's role as the most critical constituent within TELSO.

2) On the other hand, MM can enhance TELSO's performance, albeit with a less pronounced impact compared to ML. Among half of the scenarios examined, encompassing SMOP1, SMOP2, SMOP3, and SMOP8, no statistically significant alteration was observed for TELSO. Conversely, across SMOP4–7, MM statistically improved the performance of TELSO. The overall trend of the data indicates MM' s potential contribution to augmenting TELSO's effectiveness.

F. Experiments on Real-World Problems

MOEAs find diverse applications in various real-world scenarios [56]. In this study, we conducted experiments to compare the proposed TELSO with classic MOEAs in four sparse large-scale optimization fields: MOEA/PSL, PM-MOEA, S-ECSO, and SparseEA. Our objective was to evaluate the performance of TELSO on different practical problems. To this end, we selected five real-world problems. Namely, the neural network training problem (NN) [16], portfolio optimization problem (PO) [18], spark signal reconstruction problem (SR) [57], feature selection problem (FS) [58], and multi-objective knapsack problem (KP) [59]. The forms of the objective functions are shown in Table IV. Table V provides a detailed definition of these problems and the source of empirical tests. We tested the optimization ability of the algorithm on three data sets for each real-world problem to assess its effectiveness in different dimensions. Finally, we evaluated TELSO's ability to solve high-dimensional problems by conducting experiments on real-world problems with up to 5120 dimensions. For real-world problems, we set the swarm size to 500 to generate more solutions for decision-makers. All problems were set to the maximum function evaluation of 10**D*.

In this experiment, as the real PF information of the real-world problem is unknown, we used the HV [61] index to measure the algorithm's quality. The results of the HV indicators for 30 independent executions of each algorithm are presented in Table VI, displaying the average and standard deviation results. From the table, the following observations can be made.

TABLE V
DETAILED SETTINGS OF THE REAL-WORLD PROBLEMSUSED IN EXPERIMENTS

Neural network training problem	Type of variables	No. of variables	Dataset	No. of samples	No. of features	No. of classes
NN1	Real	521	Statlog_German[1]	1000	24	2
NN2		1181	Spambase[1]	4601	57	2
NN3		1241	Connectionist_bench_Sonar[1]	208	60	2
Portfolio optimization problem	**Type of variables**	**No. of variables**	**Dataset**	**No. of each instrument**	**Length of instrument**	
PO1	Real	500	EURCHF[2]	500	50	
PO2		1000		1000	50	
PO3		5000		5000	50	
Sparse signal reconstruction problem	**Type of variables**	**No. of variables**	**Dataset**	**Length of signal**	**Length of received signal**	**Sparsity of signal**
SR1	Real	1024	Synthetic[3]	1024	480	260
SR2		2048		2048	960	520
SR3		5120		5120	2400	1300
Feature selection problem	**Type of variables**	**No. of variables**	**Dataset**	**No. of samples**	**No. of features**	**No. of classes**
FS1	Binary	166	MUSK1 [17]	476	166	2
FS2		256	Semeion Handwritten Digit [17]	1593	256	10
FS3		310	LSVT Vocie Rehabilitation [17]	126	310	2
Sparse multi-objective knapsack problem	**Type of variables**	**No. of variables**	**Dataset**	**No. of items**		
KP1	Binary	1000	Synthetic [60]	1000		
KP2		2000		2000		
KP3		5000		5000		

1 http://archive.ics.uci.edu/ml
2 https://www.metatrader5.com/
3 http://www5.zzu.edu.cn/ecilab/info/1036/1011.htm

TABLE VI
HV VALUES OBTAINED BY MOEA/PSL, PM-MOEA, S-ECSO, SPARSEEA, AND TELSO ON NN, PO, SR, FS, AND KP, WHERE THE BEST RESULT IN EACH ROW IS HIGHLIGHTED. "+", "−", AND "≈" INDICATE THAT THE RESULT BY ANOTHER SLMOEA IS BETTER THAN, WORSE THAN, OR COMPARABLE TO THAT OBTAINED BY TELSO

Problem	M	D	MOEA/PSL	PM-MOEA	S-ECSO	SparseEA	TELSO
NN1	2	521	7.6807e−1 (1.69e−2)−	7.8621e−1 (6.02e−3)−	7.4462e−1 (2.76e−3)−	7.7799e−1 (6.24e−3)−	7.9630e−1 (3.93e−3)
NN2		1181	9.1490e−1 (1.30e−2)≈	9.1308e−1 (1.94e−3)−	9.0664e−1 (2.53e−3)−	9.1453e−1 (2.29e−3)≈	9.1528e−1 (2.26e−3)
NN3		1241	8.3017e−1 (1.12e−2)−	8.2513e−1 (9.09e−3)−	7.9891e−1 (8.94e−3)−	8.1965e−1 (9.61e−3)−	8.3063e−1 (8.53e−3)
PO1	2	500	1.0924e−1 (1.13e−2)−	1.2192e−1 (3.31e−3)≈	1.0240e−1 (1.15e−2)−	1.1948e−1 (3.69e−3)−	1.2298e−1 (3.00e−3)
PO2		1000	9.8151e−2 (7.78e−3)−	1.2109e−1 (3.50e−3)−	9.4549e−2 (3.12e−3)−	1.1722e−1 (2.42e−3)−	1.2234e−1 (2.29e−3)
PO3		5000	1.1955e−1 (3.73e−3)≈	1.2097e−1 (2.69e−3)≈	9.1753e−2 (3.56e−4)−	1.1991e−1 (2.96e−3)≈	1.1993e−1 (2.42e−3)
SR1	2	1024	1.5613e−1 (4.54e−2)−	1.8211e−1 (1.53e−2)−	9.6825e−2 (1.33e−3)−	1.8230e−1 (9.57e−3)−	2.8110e−1 (6.93e−3)
SR2		2048	1.1229e−1 (3.22e−2)−	1.5549e−1 (4.12e−3)−	1.0066e−1 (4.16e−3)−	1.8251e−1 (3.15e−3)−	2.4265e−1 (6.88e−3)
SR3		5120	9.2155e−2 (5.79e−2)−	1.3750e−1 (4.56e−4)−	9.9060e−2 (1.97e−3)−	1.6697e−1 (1.93e−3)−	2.1581e−1 (2.06e−3)
FS1	2	166	8.9411e−1 (8.29e−2)−	9.6474e−1 (8.81e−3)+	8.9447e−2 (1.74e−2)−	9.2711e−1 (4.29e−3)−	9.3368e−1 (8.83e−3)
FS2		256	8.4493e−1 (8.07e−2)−	9.0499e−1 (4.55e−3)+	8.6475e−2 (7.47e−2)−	8.8748e−1 (3.09e−3)≈	8.8778e−1 (2.06e−3)
FS3		310	9.8142e−1 (1.60e−2)−	9.8924e−1 (1.54e−2)≈	9.6481e−1 (4.47e−2)−	9.3451e−1 (8.03e−2)−	9.9016e−1 (1.97e−2)
KP1	2	1000	4.2834e−2 (3.87e−3)−	4.7246e−2 (1.81e−3)−	4.4785e−2 (4.48e−3)−	4.6114e−2 (1.13e−3)−	5.6596e−2 (1.37e−2)
KP2		2000	4.2487e−2 (3.74e−3)−	4.6268e−2 (1.31e−3)−	4.7987e−2 (1.78e−3)−	4.6185e−2 (7.33e−4)−	1.3347e−1 (1.34e−2)
KP3		5000	4.3977e−2 (3.38e−3)−	4.5265e−2 (1.42e−3)−	4.3548e−2 (1.47e−4)−	4.5077e−2 (5.78e−4)−	1.5490e−1 (3.18e−1)
+/−/≈			0/13/2	2/10/3	0/15/0	0/12/3	

1) The empirical results presented in the first three rows of Table VI reveal that while TELSO exhibits comparable performance to MOEA/PSL and SparseEA on the Spambase data set, it achieves the optimal values among the five algorithms

across all data sets (highlighted in the gray background), and outperforms MOEA/PSL and S-ECSO on various data sets.

2) In the fourth to sixth rows of Table VI, it is apparent that TELSO has achieved optimal values and surpassed the four most advanced sparse large-scale MOEAs in PO problems with dimensions of 500 and 1000. While PM-MOEA attained the optimal value at dimension 5000, its performance remains similar to TELSO.

3) Table VI indicates that the proposed TELSO performs remarkably well in solving the SR problem with up to 5120 dimensions, as evident from the seventh to ninth rows. Moreover, TELSO has achieved optimal results and outperformed the four SLMOEAs designed to tackle sparse problems across different dimensions.

4) TELSO outperforms MOEA/PSL, S-ECSO, and SparseEA in three different dimensions of FS and achieves the optimal values of five algorithms when the decision variables' dimension is 310. Unfortunately, when the decision variables' dimensions are 166 and 256, the proposed TELSO is inferior to PM-MOEA. However, the proposed TELSO consumes less runtime than PM-MOEA, which means that TELSO is more efficient in terms of computational runtime.

5) The last three rows of Table VI demonstrate the significant advantages of the proposed TELSO in solving KP problems. It outperforms the four SLMOEAs compared to 1000, 2000, and 5000 decision variables. It is worth mentioning that when the decision variables are further expanded to 2000 and above, TELSO's performance exhibits an overwhelming advantage.

Compared to TELSO, the statistical outcomes of the other four algorithms are 0/13/2, 2/10/3, 0/15/0, and 0/12/3, respectively. In summary, we can infer that the suggested ML can aid the swarm in finding solutions with better objective function values. In short, compared to extant SLMOEAs, TELSO performs better in nine benchmarks SMOP sets and five real-world problems.

V. Conclusion

In the present study, we first discussed the limitations of existing two-layer encoding SLMOEAs. After that, we introduce a two-layer encoding learning strategy based on frequent itemsets to solve this problem and tackle the challenge of sparse large-scale multi-objective optimization problems. Specifically, we are mining the genetic traits of multiple excellent particles. This makes inferior particles enable the collection and assimilation of advantageous evolutionary information in various examples to accelerate convergence. In addition, we propose a dynamic two-layer encoding mutation strategy tailored to the diverse demands of different evolutionary stages. This strategy assists particles in identifying more appropriate mask combinations during distinct stages of evolution.

Numerous empirical experiments on the widely adopted SMOP benchmark set have showcased the computational efficiency and effectiveness of our proposed TELSO. Compared to several state-of-the-art SLMOEAs, TELSO demonstrates superior performance and faster convergence speed. Furthermore, our optimizer performs remarkably on five real-world problems, demonstrating its capacity to handle complex real-world optimization scenarios.

In the future, we aim to explore more accurate methods to define outstanding particles better. Moreover, the proposed TELSO algorithm's performance on real-world problems requires further improvements, highlighting the necessity of designing problem-specific algorithms that better address the unique landscape of diverse real-world optimization scenarios.

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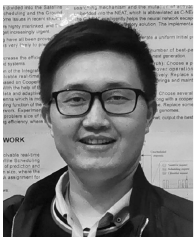
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